Communication

Two everyday uses of satellite technology are telecommunications and navigation. The location of satellites high above Earth makes them ideal for sending and receiving radio signals between widely separated regions. Signals from a network of satellites also allow navigation systems to pinpoint our position on the ground and show us the best route to our destination.

FAST FACTS

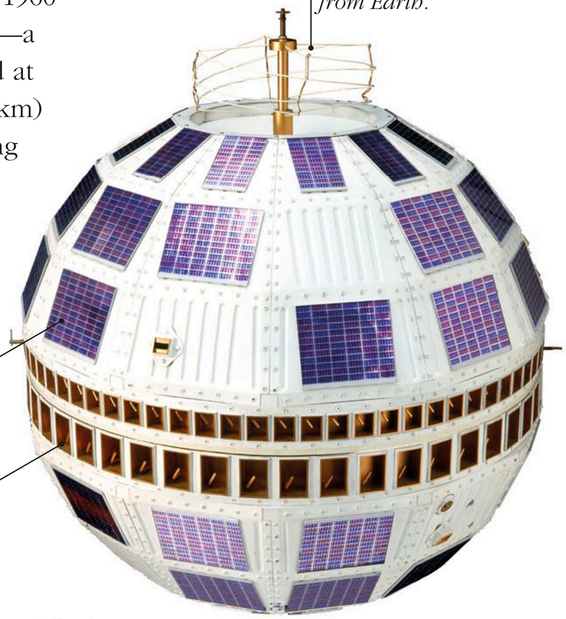
- Comsats (communication satellites) relay signals for cell phones, the Internet, podcasts, TV, radio, and more.
- Geostationary satellites orbit Earth above the equator and cannot relay signals near the poles, so comsats for high-latitude regions follow elliptical orbits.
- Satnav (satellite navigation) systems rely on signals from dozens of satellites in orbit.

Total mass of satellite is just 400 lb (180 kg)

Inflated satellite is 100 ft (30.4 m) in diameter

BOUNCING RADIO SIGNALS

The simplest way of communicating via satellite is to use a reflector that simply bounces a beam of radio waves back to Earth. NASA's Echo 1A satellite of 1960 was an early test of this principle—a huge metallic balloon that orbited at heights of up to 994 miles (1,600 km) and provided a target for bouncing radio signals.



Curved antennae receive satellite commands from Earth.

Solar cells generate 14 watts of power—less than a light bulb, but enough to run the satellite.

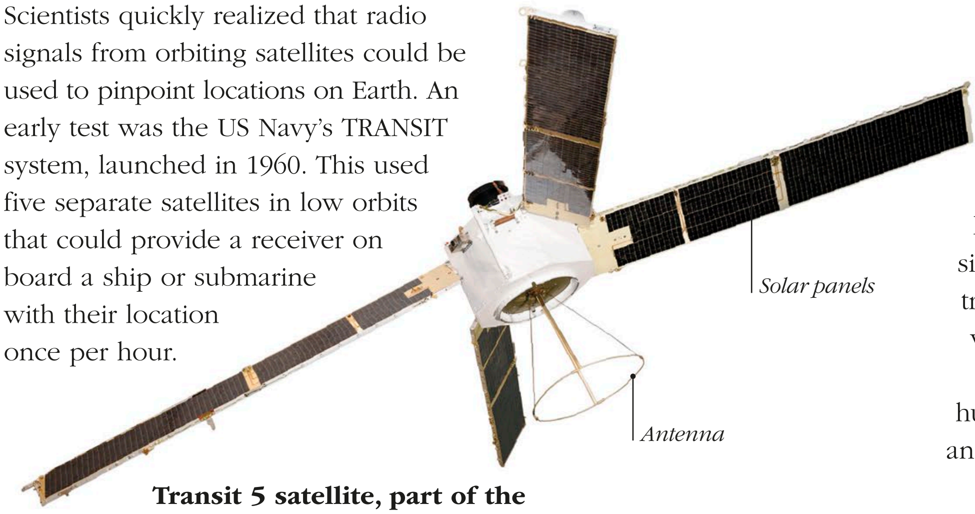
Equatorial antennae relay signals and keep Earth in view as the satellite spins.

COMMUNICATIONS SATELLITE

Active communications satellites have electronics on board to receive signals from one antenna on Earth and transmit them toward another. Telstar 1 was the first of these "comsats" to be launched. It entered low orbit a few hundred miles above Earth in July 1962, and carried the first television broadcasts across the Atlantic Ocean.

SATELLITE NAVIGATION

Scientists quickly realized that radio signals from orbiting satellites could be used to pinpoint locations on Earth. An early test was the US Navy's TRANSIT system, launched in 1960. This used five separate satellites in low orbits that could provide a receiver on board a ship or submarine with their location once per hour.



Solar panels

Antenna

Transit 5 satellite, part of the first satellite navigation system